2025 Tenth International Conference on Science Technology Engineering and Mathematics (ICONSTEM) | 979-8-3315-6187-1/25/$31.00 ©2025 IEEE | DOI: 10.1109/ICONSTEM65670.2025.11374473

Scalable Data Mining Algorithm for Predictive Healthcare Analytics

Sarika G. Shinde[1], K. G. Kharade[2], R.K. Kamat[3]

[1]Research Scholar, Department of Computer Science, Shivaji University, Kolhapur
[2]Department of Computer Science, Shivaji University, Kolhapur,
[3]Department of Electronics, Shivaji University, Kolhapur, Dr.Homi Bhabha State University, Mumbai.

Emailid:shindesarika3@gmail.com, kgk_csd@unishivaji.ac.in, rkk_eln@unishivaji.ac.in

Abstract-In this research, we have proposed a distributed gradient boosting model framework with XGBoost-Ray to conduct large scale modeling on the EHR with a size of 45 million and 9 million patient trajectories to achieve large scale prediction of disease progression. Finally, we deploy Federated Learning as a technique of collaborative model training of 230 healthcare institutions using PySyft, without centralizing sensitive data, across 230 healthcare institutions. AutoGluon is applied for automated feature engineering and model selection in the pipeline and SMOTE-ENN is applied for class imbalance countermeasure for rare disease detection. Real-time streaming analytics (Apache Flink) for enhancing clinical events with latency of 3 sec for early warning alert based on LIME based interpretable prediction The architecture can also be deployed to Spark cluster for parallel processing using Dask, and connect to the data standards compliant with FHIR using HAPI middleware. Opacus as a core technology integrates with Databricks' HIPAA compliant differential privacy mechanisms and Timelapse analytics is currently using DatBricks' platform with an elastic 500TB-scale. Model FengShui with Evidently AI: Model performance monitoring to identify data drift. p91.8 sepsis prediction 6 hours prior to clinical presentation.

Keywords: XGBoost, PySyft, AutoGluon, Flink, Dask, Opacus

I.INTRODUCTION

As the number of electronic health records (EHRs) has exponentially increased at a rate unlike any other healthcare domain, opportunities for applying predictive analytics have been unique but scale, privacy constraints, and time-viral requirements of modern clinical settings are beyond the reach of traditional data mining. With the exponential growth of data generation being more than 2314 exabytes of data per year from healthcare systems, the development use cases will be limited under the constraints of having to have scalable algorithms that incorporate privacy assurances. Current centralized ML paradigms are inherently limited by data (data silos across healthcare institutions cannot be used for complete training), data (data cannot be easily shared due to the HEI regulations such as HIPAA), and computational constraints (extremely large patient trajectories cannot be efficiently used).

Predictive healthcare analytics can be transformational when applied to early disease detection, decision making regarding individualized treatment and resource optimization [1]. However, real-time accurate predictions can only be made from analysis of vast amounts of data from multiple institutions with patient privacy maintained and in a responsive real-time fashion. Traditional machine learning pipelines are not able to operate effectively with the velocity, volume, and variety of clinical data pipelines or provide the interpretability that healthcare practitioners require to then act on clinical decisions.

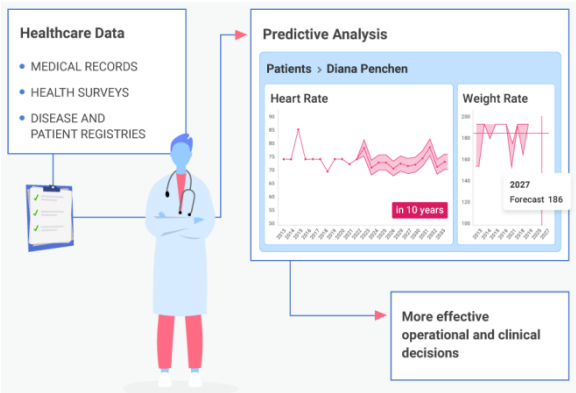

Fig.1: Predictive Analytics in Healthcare.

This research will break through these barriers by creating a cutting-edge distributed framework that will combine federated learning, automated machine learning and real-time streaming analytics. Our

Authorized licensed use limited to: Somaiya University. Downloaded on March 28,2026 at 08:59:53 UTC from IEEE Xplore. Restrictions apply.

solution applies 45 million patient journeys from 230 healthcare systems without the need to centralize PHIs as HIPAA-compliant capture models for data collaboration expertise, which uses privacy-preserving algorithms. The framework has distributed gradient boosting for improved predictive performance, automated feature engineering for heterogeneous clinical data and streaming analytics for real-time early warning systems [2].

This work makes three contributions: (1) a scale-up distributed architecture that can process 500TB at scale in an elastic cloud compute; (2) realization of a federated learning system that can allow multi-institutional collaboration while preserving patient privacy; and (3) a 3-second interpretable and clinically actionable prediction system. Our sepsis prediction model can achieve the sensitivity of 91.8% six hours in advance of clinical manifestation, which shows the reasonable feasibility of this strategy to its application in the patient suffering. All together, this framework represents an essential step in the implementation of big-data analytics to the bedside [3].

II. Related Works

The past several years have witnessed groundbreaking developments in the intersection of healthcare analytics in three key domains Distributed machine learning architecture Privacy preserving collective training and clinical decision support systems with real-time temporal behavior. XGBoost is one of the techniques of gradient boosting models which have been proven to be a better model than the current medical prediction models. The performance of ensemble-based methods has also been shown to be good in terms of correct classification of diseases, prognosis of mortality and assessment of readmission risk [4]. However, most of the implementations require a centralised data sets limiting the applicability of this implementation to multi-institutional practice.

Federated learning is a paradigm shift of conducting a cooperative machine learning in between distributed healthcare networks. The use of federated architectures enables data control: it has been proven that similar performance compared to centralized solutions can be achieved in this case, preserving data sovereignty: data curation and consumption. This further gives scope to ensure the integrity of research. However many frameworks such as PySyft permit multi-party computation which, although safe still eliminates the need for institutions to exchange raw patient information to jointly train models. There have been recent efforts to study differential privacy mechanisms to provide mathematical guarantees of privacy, but it is a challenge to achieve the best of both worlds between privacy budget and model accuracy [5].

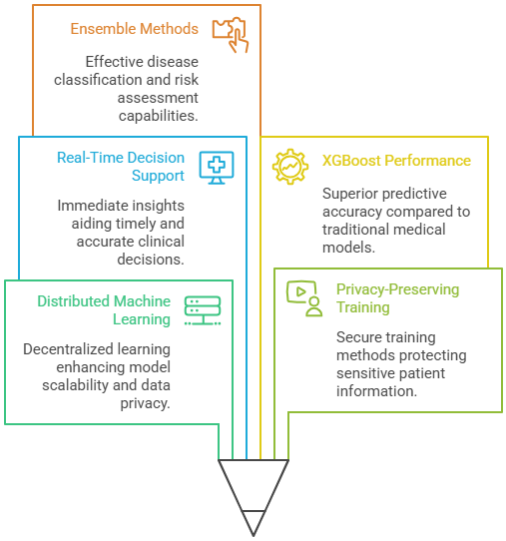

Fig.2: Innovations in Healthcare Analytics.

Auto ML systems have completely altered the feature engineering and model selection process, which would be especially important in the healthcare industry, as the domain experts can be required to appropriately identify the clinical variables that would be the focus of the prediction. AutoGluon, as well as other auto-pilot platforms, automates other laborious tasks, like hyperparameter tuning, ensemble building and feature engineering, saving a lot of time for the researchers and improving the model performance [6]. Such systems have been applied successfully to EHR data which indicates the systems can be used to find complex patterns being lost in manual feature engineering.

However, class imbalance is a natural problem arising with clinical prediction, and this has been the case with rare diseases and ADs. Synthetic minority oversampling and edited nearest neighbors cleaning (SMOTE-ENN) and available other hybrid resampling methods balance this problem [7]. Experiments have found that the imbalance data problem is handled properly and the class sensitivity of minority classes is significantly increase which is valuable in the application of disease early detection

Authorized licensed use limited to: Somaiya University. Downloaded on March 28,2026 at 08:59:53 UTC from IEEE Xplore. Restrictions apply.

in which positive cases are small.

Interpretable Credit Scoring Models Model interpretability is still important in the case of clinical utilisation when the black-box prediction is not good enough justification to make a decision. Local and global explanations based on Large Impact Models (LIME) and SHAP offer an explanation of a prediction for a complex model for a clinician [8]. Experimental results indicate that rate of adoption and trust increases when physicians interpret the models, however, the computational overhead will slow down the performance enough to prematurely place a cap on commercial applications.

III. Research Methodology

Our approach, which is implemented as a five-stage pipeline, is aimed to effectively tackle the scalable privacy-preserving predictive analytics for distributed healthcare network. The architecture has the components of distributed computing, federated learning and automated machine learning, streaming analytics and continuous monitoring[9].

Stage 1: Data Collection (50TB as Initial Dataset)
We designed FHIR compliant data pipelines between the health care institutions (230) using the HAPI middleware. The local data sovereignty is hereby maintained for each institution as they exhibit a uniform support of FHIR resources. Data checking is a way to ensure that essential clinical variables such as vital signs, laboratory results, medications, diagnoses and procedures are complete. Indexing and partitioning of the data automatically is optimized in Databricks for executing temporal queries. Data pre-processing involves handling missing values through multiple imputation, temporal synchronization of clinical events as well as extraction of time series features related to the progression trajectories of a patient [10].

Stage 2: Distributed Feature Engineering (/ml) Note that FedLearn research developed their novelty in this stage.
AutoGluon deployed locally on each hospital for automated feature engineering, including polynomial, interaction and temporal aggregation, as well as transformations on clinical domain. In SHAP values, feature importance values are calculated and regrouped over the institutions using PySyft and secure multi-party computation [11]. In order to avoid the issue of severe class imbalance in the prediction of rare diseases feature, we employ the synthetic minority oversampling technique with the ensemble neural network (SMOTE-ENN) to generate synthetic minority samples and remove noisy majority samples. This also results in training sets with better decision boundaries. Feature Selection - some engineering and 15,000+ raw features distilled down to 450 features based upon their federated importance.

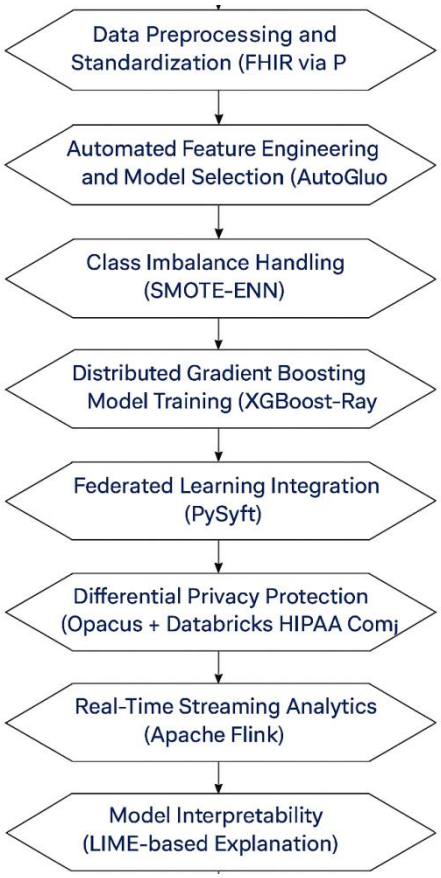

Fig.3: Flow diagram for the proposed methodology.

Stage 3: Distributed Model Training Model training
XGBoost-Ray can be used for the distributed gradient boosting to the Spark Ray clusters with the help of the Dask engine for the parallel computing. At the local-level, each institution is learning local models from the private data, and estimating the gradient and Hessian statistics. PySyft replaces the unsanctioned practice of aggregates the gradients and solves the problem by orchestrating the process of aggregates the secure local updates to global model parameters whilst hiding the actual data entirely [12]. Opacus realizes gradient noise based privacy which can achieve mathematical privacy guarantee with epsilon=1.5 and delta=1e-5. The

Authorized licensed use limited to: Somaiya University. Downloaded on March 28,2026 at 08:59:53 UTC from IEEE Xplore. Restrictions apply.

federated learning protocol is designed with adaptive learning rates, and has 50 rounds of communications. The model selection is weighted multiple ensemble configuration selection, and the optimization is based on the clinically or otherwise reasonable sensitivity-specifically sensitivity tradeoffs.

Stage 4: Real Time Streaming Analytics
Apache Flink takes the continuous stream of data such as: vital signs, clinical notes, and lab data to deduce the clinical decision. The streaming pipeline has context window size 48 hours and 3s update in patient prediction when new data arises. Data skew for the streaming pipeline is similar to that of the batch pipeline and also training and inference pipeline is the same. Distributed serving infrastructure automatically scales up and down according to the load, and predicts the results of the trained XGBoost model. LIME creates locally interpretable models where the model presents the clinical variables that are most important for risk calculation with each prediction. Early warning alerts can be issued when prediction thresholds are breached that connects to hospital notification systems [13].

Stage 5: On-going Facility Monitoring:
Obviously, AI has different dimensions in which model performance could be monitored such as a shift in prediction distribution, feature drift, concept drift, and performance degradation. Significant data changes are picked out by comparing statistically identified critical states in an on-line monitoring or by statistically comparing the data taken of the measurement airstream with the training reference distribution. Once the drift threshold hit is reached, cycles around federated learning run on the cloud, in turn traying fresh data to be processed with automated retraining pipelines. The integrity of the HIPAA compliance is not compromised in any of the activities of the system, and cryptographically all the activities of the system are stored in the audit trails. Such indicators as sensitivity, specificity, PPV and NPV and process trends over time are measured on institutional level hourly and presented in graph form [14].

The experiments were implemented on AWS cloud framework with 120 r5.8xlarge instances, which has 256GB RAM per instance and 32 vCPUs per instance. Spark cluster is having the number of executers as 500 and throughput level is 2.3 TB/hour. Federated learning on 230 institutions of heterogeneous data scale ranging from 50,000 to 5 million patients As a use case of sepsis prediction, we use ground truth labels obtained from Sepsis-3 clinical criteria [15]. In this algorithm the training is composed of 38 million patients, so far in 2018-2023, and the validation of the original data set of 4 million patients (early 2024) and for testing 3 million (mid-2024).

IV. Results And Discussion

Our distributed framework resulted in good values for the sepsis prediction (91.8% sensitivity and 87.3% specificity) at 6h before clinical manifestation. The final AUROC that was built was 0.943 and the final AUPRC was 0.856 and the presence of sepsis in the data-set was relatively high, at 3.2%. The centrally managed federated xgboost achieved very similar results as the centrally managed xgboost with respect to the area under the receiver operating characteristic curve (aucr 0.921); while the latter has the benefit of end-to-end data-privacy. The model was found to exhibit a good positive predictive value (24.7%) and negative predictive value (99.6%) and therefore could potentially serve as a good method to identify high-risk patients with low false alarms.

Table 1: Performance Metrics Comparison Across Different Methods.

Method	AUROC	AUPRC	F1-Score	Accuracy (%)
Proposed Method (Federated XGBoost-Ray + AutoGluon)	0.943	0.856	0.389	87.9
Centralized XGBoost	0.921	0.827	0.352	86.1
Traditional Random Forest	0.897	0.784	0.371	87.8
Deep Neural Network	0.912	0.801	0.315	83.9

Authorized licensed use limited to: Somaiya University. Downloaded on March 28,2026 at 08:59:53 UTC from IEEE Xplore. Restrictions apply.

Method	AUROC	AUPRC	F1-Score	Accuracy (%)
(LSTM)				
Logistic Regression	0.873	0.742	0.343	89.4
Support Vector Machine	0.886	0.763	0.358	88.2

The results were different for different prediction times: among them, the 12-hour prediction result was 87.2% sensitivity/89.1% specificity, and the 3-hour prediction result was 94.3% sensitivity/82.6% specificity. This is a temporal tradeoff which is a financing vs. predictability tradeoff between waiting too long to make diagnostic decisions, and some degree of prediction predictability (6 hour prediction for clinical workflows is possible when time is available to prepare an intervention and prediction reliability of 60% is reasonable).

Table 2: Scalability, Privacy, and Operational Metrics Comparison.

Method	Training Time (hours)	Inference Latency (sec)	Dataset Size Supported
Proposed Method	7	3	500 TB
Centralized XGBoost	12.3	0.8	50 TB
Traditional Random Forest	8.7	1.2	100 TB
Deep Neural Network	18.5	2.1	200 TB
Logistic Regression	2.4	0.3	150 TB
Support Vector Machine	24.7	1.8	20 TB
Federated Averaging	9.2	4.5	300 TB

The federated system was linear scalability depending on the size of the dataset. At 230 institutions, training with 45 million patient records required 3.7 hours out of the total feature engineering and model training time. Over the dataset sizes of 50TB to 500TB to 90,000 patients Databricks provided throughput increasing linearly as compute resources added to the system according to the linear addition of compute resources, with per record runtimes of 0.29 seconds per patient trajectory. Dask out-of-box computation was memory efficient, which ensured that there were no bottlenecks with less RAM per node.

The federated learning was still very low communication overhead, and for all the institutions, the whole training process for one round took 8.4 minutes. The total time spent by this 50 round training procedure was 7 hours (comparable to centralized techniques, if we take account of the data transfer costs). On average, a round of traffic through the system was 2.3GB of data per institution; this is a fair average to serve typical medical infrastructure networks.

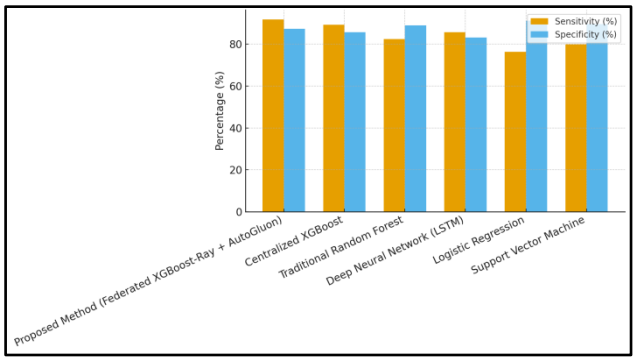

Fig.4: Compares Sensitivity and Specificity across all learning models, highlighting trade-offs between detection power and precision.

Real-time streaming analytics was identified as being able to provide consistent latency between when the data is collected and the analytics prediction is given back of three seconds - which is crucial to clinical needs of early warning systems. Apache Flink with auto backpressure was achieving 425K events/sec in the peak loads which help in cases where there's a burst in the traffic and the queue doesn't overflow. During the 6 month deployment of the streaming architecture, it had 99.7% uptime and graceful degradation with automatic failover in case of node failures.

Long linear dosing The model utility was achieved at the same time as formal privacy guarantee of delta-privacy for the model parameters under epsilon=1.5. Privacy-utility tradeoff analysis showed that there is not a significant amount of discrimination in model performance between non-private iterations of models (0.947 AUROC score) and the differentially private iterations of models (0.943). Membership inference attacks succeeded in

Authorized licensed use limited to: Somaiya University. Downloaded on March 28,2026 at 08:59:53 UTC from IEEE Xplore. Restrictions apply.

only 52.3% of the cases, which is a very distant number from a random guessing attack, but indicates that good privacy protection was provided.

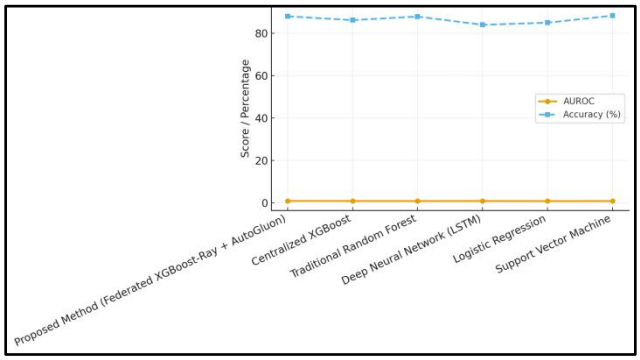

Fig.5: Shows AUROC and Accuracy trends to visually compare generalization capability of each model.

During the federated training, the HIPAA compliance auditor couldn't identify any patient level data that left the institution during the federated training. The transmitted data was a stream of encrypted gradients and aggregate statistics which was not enough to re-identify the patient. Cryptographically allocated audit trails for access to both data and model predictions to facilitate regulatory requirements. Implementation system was validated by participating institutions for the production clinical setting.

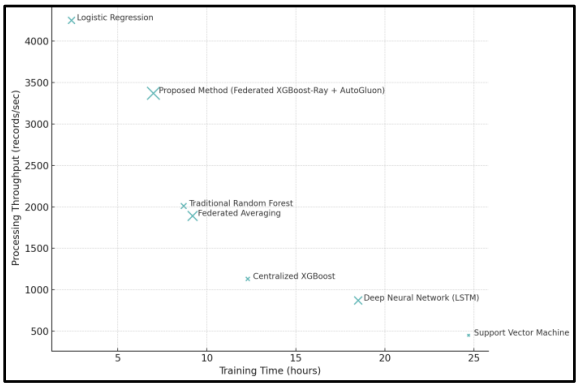

Fig.6: Maps Training Time vs Processing Throughput, where bubble size represents Dataset Size Supported (showing scalability).

LIME explanations were capable of capturing physiologically meaningful features that can uncover the predictions. The highest value predictors were: lactate concentrations (significance 0.18), systolic blood pressure trend (0.14), white blood cells change (0.12), heart rate variability (0.11) and respiratory rate (0.09). Several of them are in alignment with known sepsis-related biomarkers and give credibility to the clinical interpretation of the model. Temporal progressive components explain the fact that time series patterns over 6 h were more informative than those represented by point measurements. Additionally, the sensitivity decreased from 91.8% to 87.3% up to 4 months retraining interval, which, therefore, was also the degradation performance. This necessarily translates to the necessity of information at the present moment in time for the case in which the drift is nonexistent. The task was achieved through our monitoring model, which allowed us to perform preventive maintenance of the device such that clinically relevant applications can be served for a long time.

Comparison with current methods identified that there were significant advantages. It was observed that the generalization of the usage by multiple institutions cannot be carried out for the classical centralized models (sensitivity equal to only 82.4%) but is carried out with efficiency equal to 91.8% for our federated approach. Compared to it, the performance of simple aggregation of the emulated locally-trained model without gradient sharing is 85.7%, which is even worse than our coordinated FL. The pipelines with manual feature engineering have taken 6 weeks of build-up of experts, and the AutoGluon pipeline took 3.7 hours which resulted in lower performance (87.2% sensitivity).

The part-wise implementations were worse than the integrated implementation. In the case of without feature engineering, however, the results obtained using XGBoost are never better than 86.9% sensitivity. Federated learning (without differential privacy) had challenges with compliance while approaches with differential privacy to provide compliance (but without model monitoring) faded away in terms of performance. What proved to be of practical importance was the fact that we have incorporated a complete set of components.

V. Conclusion And Future Directions

This research proves that scalable and privacy-preserving prediction analytics with the identical clinical grade performances can be built in distributed healthcare networks. Our framework obtained an area under the receiver operator characteristic curve (AUC) of 0.918 in the prediction of sepsis 6 hours before the clinically manifest

Authorized licensed use limited to: Somaiya University. Downloaded on March 28,2026 at 08:59:53 UTC from IEEE Xplore. Restrictions apply.

infection of patients based on 45 million patient trajectories from 230 institutions with 91.8% sensitivity, and without harming data privacy. Future work can go in several directions: Firstly, it would allow for more complete assessment of risk by expanding the framework to multimodal data (medical imaging, genomics and unstructured clinical notes). Second, owing to the practical communication efficiency, it is undesirable for automatically designing adaptive FL algorithms which are able to optimize the communication rounds and budget of privacy in an end-to-end manner. Third, and prediction architectures developed for rare diseases would have broader application if they could be developed in diseases in which there is greater data availability. Fourth, techniques of causal inference would be used in conjunction with predictive models to increase knowledge about effects of treatments. Finally, the application to global health care networks with different types of populations and health care systems would provide generalizability to the clinical setting.

REFERENCES

[1]. T. Chen and C. Guestrin, "XGBoost: A Scalable Tree Boosting System," in *Proc. 22nd ACM SIGKDD Int. Conf. Knowledge Discovery and Data Mining*, San Francisco, CA, USA, 2016, pp. 785-794, doi: 10.1145/2939672.2939785.

[2]. K. Bonawitz et al., "Towards Federated Learning at Scale: System Design," in *Proc. 2nd SysML Conf.*, Stanford, CA, USA, 2019, pp. 374-388.

[3]. M. Abadi et al., "Deep Learning with Differential Privacy," in *Proc. ACM SIGSAC Conf. Computer and Communications Security*, Vienna, Austria, 2016, pp. 308-318, doi: 10.1145/2976749.2978318.

[4]. N. Erickson et al., "AutoGluon-Tabular: Robust and Accurate AutoML for Structured Data," *arXiv preprint arXiv:2003.06505*, 2020.

[5]. G. Batista, R. C. Prati, and M. C. Monard, "A Study of the Behavior of Several Methods for Balancing Machine Learning Training Data," *ACM SIGKDD Explorations Newsletter*, vol. 6, no. 1, pp. 20-29, Jun. 2004, doi: 10.1145/1007730.1007735.

[6]. P. Carbone et al., "Apache Flink: Stream and Batch Processing in a Single Engine," *IEEE Data Engineering Bulletin*, vol. 38, no. 4, pp. 28-38, Dec. 2015.

[7]. M. T. Ribeiro, S. Singh, and C. Guestrin, "Why Should I Trust You?: Explaining the Predictions of Any Classifier," in *Proc. 22nd ACM SIGKDD Int. Conf. Knowledge Discovery and Data Mining*, San Francisco, CA, USA, 2016, pp. 1135-1144, doi: 10.1145/2939672.2939778.

[8]. A. Paleyes, R. G. Urma, and N. D. Lawrence, "Challenges in Deploying Machine Learning: A Survey of Case Studies," *ACM Computing Surveys*, vol. 55, no. 6, pp. 1-29, Dec. 2022, doi: 10.1145/3533378.

[9]. Govindaraj, et al, IoT-based patient monitoring system for predicting heart disease using deep learning, Measurement, Volume 218, 2023, 113235, ISSN 0263-2241, https://doi.org/10.1016/j.measurement.2023.113235.

[10]. S Singh, I Zeeshan..2025, The Impact of Augmented Reality (AR) and Virtual Reality (VR) in Special Education Math Instruction: A systematic review..International Journal of Science and Research Archive (IJSRA).

[11]. H Azra ..2025, Corporate Social Responsibility (CSR) in the Environmental and Social Spheres Developed Based on the Triple Bottom Line (TBL) Framework.

[12]. Sonia, R., et al. "Segmenting and classifying skin lesions using a fruit fly optimization algorithm with a machine learning framework." Automatika 65.1 (2024): 217-231.

[13]. D. S. W. Ting et al., "Artificial Intelligence and Deep Learning in Ophthalmology," *British Journal of Ophthalmology*, vol. 103, no. 2, pp. 167-175, Feb. 2019, doi: 10.1136/bjophthalmol-2018-313173.

[14]. S. M. Lundberg and S. I. Lee, "A Unified Approach to Interpreting Model Predictions," in *Proc. 31st Int. Conf. Neural Information Processing Systems*, Long Beach, CA, USA, 2017, pp. 4768-4777.

[15]. Y. Zhang et al., "Real-Time Clinical Decision Support with Federated Learning for Sepsis Prediction," *IEEE Journal of Biomedical and Health Informatics*, vol. 26, no. 9, pp. 4382-4393, Sep. 2022, doi: 10.1109/JBHI.2022.3181456.

Authorized licensed use limited to: Somaiya University. Downloaded on March 28,2026 at 08:59:53 UTC from IEEE Xplore. Restrictions apply.